

7. LITERATURE REVIEW

Human–Computer Interaction (HCI) has evolved significantly with the integration of Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision technologies. Among various interaction methods, hand gesture recognition has emerged as an effective contactless approach that enables users to interact with digital systems using natural hand movements. Recent research has focused on improving gesture recognition accuracy, reducing computational complexity, and enabling real-time applications. The following literature review summarizes the major contributions of previous studies relevant to the proposed project.

Meng et al. (2024) proposed a real-time hand gesture monitoring model based on registerable gesture recognition for accurate hand gesture detection. The system demonstrated high recognition accuracy and robustness under controlled conditions using AI-based techniques. However, the work primarily focused on gesture recognition and monitoring without extending its application to practical systems such as presentation control. This creates an opportunity to develop application-oriented solutions that utilize gesture recognition for real-world tasks.

Yaseen et al. (2024) developed a next-generation dynamic hand gesture recognition system by combining MediaPipe with deep learning techniques for Human–Computer Interaction. Their model effectively extracted hand landmarks and classified dynamic gestures with high accuracy. Although the proposed system achieved promising performance, it did not provide an integrated platform for presentation management or user interaction, leaving scope for practical implementations in educational and corporate environments.

Kumar et al. (2023) developed a real-time American Sign Language (ASL) recognition system using MediaPipe and Convolutional Neural Networks (CNNs). Their approach demonstrated that combining hand landmark detection with deep learning significantly improves gesture classification accuracy. However, the research focused specifically on sign language interpretation rather than gesture-controlled application interfaces such as presentation systems.

RESEARCH GAP

Based on the review of existing research on hand gesture recognition and vision-based Human–Computer Interaction systems, the following research gaps have been identified:

1. Lack of Complete Presentation Control Systems

Most existing studies focus primarily on hand gesture detection and recognition. However, they do not provide a complete presentation control solution that integrates features such as slide navigation, virtual pointer control, annotation, and presentation management into a single system.

2. Dependence on Specialized or Expensive Hardware

Many gesture recognition systems require dedicated hardware such as Kinect sensors, infrared cameras, depth sensors, or sensor gloves. These hardware requirements increase implementation costs and reduce the practicality and accessibility of the system for everyday users.

3. Limited Use of Lightweight and Real-Time Solutions

Several state-of-the-art approaches rely on computationally intensive deep learning models that require high-end hardware for real-time performance. There is a need for a lightweight solution that provides accurate gesture recognition using only a standard webcam while maintaining low latency.

4. Insufficient Integration with Modern Web Technologies

Most existing implementations are standalone applications or research prototypes with limited scalability. Very few systems integrate computer vision with modern web technologies such as Flask, React, and MongoDB to create a complete, user-friendly, and easily deployable application.

5. Limited User Experience and Practical Functionality

Existing systems mainly demonstrate gesture recognition capabilities but often lack practical features such as user profile management, customized gesture preferences, virtual annotations, and seamless interaction during presentations. There is a need for a comprehensive system that enhances presenter mobility, audience engagement, and overall presentation efficiency.

Proposed Contribution

The proposed project, **VisionX: AI-Powered Vision-Based Intelligent Presentation Control System**, aims to provide an intelligent, real-time, and contactless presentation control solution by integrating Artificial Intelligence, Computer Vision, and modern web technologies. The major contributions of the proposed system are as follows:

- 1. Development of a Contactless Presentation Control System**
- 2. Real-Time Hand Gesture Recognition**
- 3. AI-Based Human–Computer Interaction**
- 4. Intelligent Presentation Navigation**
- 5. Virtual Pointer and Annotation Features**

8. REFERENCES

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Key problems that are been solved –

1. Dependence on Physical Presentation Devices

Eliminates the need for keyboards, mice, wireless clickers, or presentation remotes by enabling gesture-based slide control.

2. Limited Presenter Mobility

Allows presenters to move freely while delivering presentations without being restricted to a podium or input device.

3. High Cost of Existing Gesture-Control Systems

Replaces expensive hardware such as sensor gloves, Kinect, or depth cameras with a standard webcam and computer vision techniques.

4. Interrupted Presentation Flow

Enables natural hand gestures for slide navigation, virtual pointer control, and annotations, reducing distractions caused by manual interaction with devices.

5. Lack of Contactless Human–Computer Interaction

Provides a touch-free, AI-powered presentation control system using real-time hand gesture recognition, making presentations more intuitive, hygienic, and user-friendly.